

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY
COURSE CODE: CSA5117

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

QUESTIONS:

Total Marks: 50

Annexure A

Code of Conduct:

I Harshini R N (192421175) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Scenario-based

1. Secure Data Transmission using Block Cipher

A company is transmitting repeated financial data blocks over a network. Initially, they used ECB mode and noticed patterns in ciphertext.

- A. Explain why this issue occurs.
- B. Suggest a more secure block cipher mode and justify your choice based on the scenario.

2. Choosing an Encryption Algorithm for Banking System

A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish.

- A. Identify the most suitable algorithm for this scenario.
- B. Justify your choice based on security strength and efficiency.

3. Key Exchange in an Unsecured Network

Two users want to establish a secure communication channel over a public network using Diffie-Hellman key exchange. However, the network is vulnerable to attacks.

- A. Explain how the key exchange works in this scenario.
- B. Identify a possible security threat and suggest a solution to mitigate it.

4. Public Key Cryptography for Email Security

An organization plans to implement RSA for secure email communication.

- A. Describe how RSA can be used for encryption and decryption in this scenario.
- B. Explain how key distribution and management play a role in maintaining security.

5. Cryptography for IoT Devices

A smart home system uses low-power IoT devices that need secure communication. The designer must choose between RSA and Elliptic Curve Cryptography (ECC).

- A. Analyze the constraints of IoT devices.
- B. Recommend a suitable cryptographic approach and justify your answer.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

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ASSESSMENT TOOL 2

Scenario-based

Name: Harshini R N

Reg No: 192421175

COURSE NAME: Cryptography and Network Security

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1. Secure Data Transmission using Block Cipher

A. Explain why this issue occurs.

The issue occurs because the Electronic Codebook (ECB) mode encrypts identical plaintext blocks into identical ciphertext blocks. Since financial data often contains highly structured and repetitive information (like padded zeros, standard headers, or repeated transaction formats), ECB preserves these underlying data patterns in the ciphertext. This allows attackers to perform pattern analysis and potentially deduce the contents without needing the encryption key.

B. Suggest a more secure block cipher mode and justify your choice.

Cipher Block Chaining (CBC) or **Galois/Counter Mode (GCM)** are significantly more secure.

- **Justification:** CBC uses an Initialization Vector (IV) and XORs each plaintext block with the previous ciphertext block before encryption. This randomized chaining means that even if two plaintext blocks are identical, their resulting ciphertext blocks will be completely different, eliminating the pattern vulnerability of ECB. For financial data, **GCM** is often the gold standard because it provides both data confidentiality and authentication (ensuring the data hasn't been tampered with in transit).

2. Choosing an Encryption Algorithm for Banking System

A. Identify the most suitable algorithm for this scenario.

Among the provided options (DES, 3-DES, and Blowfish), **Blowfish** is the most suitable algorithm.

B. Justify your choice based on security strength and efficiency.

- **DES** is obsolete and highly insecure due to its small 56-bit key size, which can be easily broken via brute-force attacks.

- **3-DES** (Triple DES) provides adequate security by applying the DES algorithm three times, but it is notoriously slow and computationally heavy, which violates the "moderate performance" requirement of the banking system.
- **Blowfish** strikes the best balance. It offers a highly variable key length (up to 448 bits), making it highly secure against modern attacks, and it is significantly faster and more efficient in software execution than 3-DES.

3. Key Exchange in an Unsecured Network

A. Explain how the key exchange works in this scenario.

The Diffie-Hellman (DH) key exchange allows two parties to generate a shared secret over an insecure channel without ever transmitting the secret itself.

1. Both users agree on public parameters: a large prime number p and a generator base g .
2. User A selects a private secret a and calculates their public key: $A = g^a \pmod{p}$.
3. User B selects a private secret b and calculates their public key: $B = g^b \pmod{p}$.
4. They exchange public keys A and B over the network.
5. User A calculates the shared secret: $S = B^a \pmod{p}$.
6. User B calculates the same shared secret: $S = A^b \pmod{p}$. Both now share S for symmetric encryption.

B. Identify a possible security threat and suggest a solution to mitigate it.

- **Threat:** The primary threat is a **Man-in-the-Middle (MitM) attack**. Because basic Diffie-Hellman does not authenticate the users, an attacker can intercept the public keys in transit, negotiate separate shared secrets with User A and User B, and then secretly decrypt, read, and re-encrypt the traffic passing between them.
- **Solution:** Implement **Authenticated Diffie-Hellman**. This is done by combining DH with digital signatures (like RSA or ECDSA) or digital certificates issued by a Public Key Infrastructure (PKI). This ensures that before the key exchange is finalized, both parties can mathematically verify the exact identity of the person sending the public key.

4. Public Key Cryptography for Email Security

A. Describe how RSA can be used for encryption and decryption in this scenario.

In RSA email encryption, every user has a key pair: a public key (available to anyone) and a private key (kept strictly secret).

- **Encryption:** When User A wants to send a secure email to User B, User A obtains User B's public key. User A encrypts the email using User B's public key.
- **Decryption:** Once encrypted, the ciphertext can *only* be decrypted by the mathematically corresponding private key. When User B receives the email, they use their own private key to decrypt and read it.

B. Explain how key distribution and management play a role in maintaining security.

RSA's security relies entirely on the trust that a public key actually belongs to the intended recipient. If an attacker successfully distributes their own public key while pretending to be User B, the sender will unknowingly encrypt the email for the attacker.

To manage this, organizations use a **Public Key Infrastructure (PKI)** and a **Certificate Authority (CA)**. The CA verifies user identities and issues digital certificates binding a user to their public key. Key management also involves strictly safeguarding the private keys and having revocation protocols (CRLs) in case a private key is compromised.

5. Cryptography for IoT Devices

A. Analyze the constraints of IoT devices.

Smart home IoT devices (like sensors, smart locks, or thermostats) are highly constrained environments. They typically feature low-power microcontrollers, limited memory (RAM/ROM), rely on battery power, and communicate over low-bandwidth networks. Cryptographic operations that require heavy mathematical computation drain batteries quickly and cause unacceptable network latency.

B. Recommend a suitable cryptographic approach and justify your answer.

Elliptic Curve Cryptography (ECC) is the clear recommendation.

- **Justification:** ECC provides the exact same level of cryptographic security as RSA but uses vastly smaller key sizes. For example, a 256-bit ECC key offers comparable security to a 3072-bit RSA key. Because the keys are smaller, the mathematical operations require significantly less computational power, consume less memory, transmit faster over low-bandwidth networks, and critically, preserve the battery life of the IoT devices.